Loss of heterozygosity in malignant gliomas involves at least three distinct regions on chromosome 10.

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A panel of glial tumors consisting of 11 low grade gliomas, 9 anaplastic gliomas, and 29 glioblastomas were analyzed for loss of heterozygosity by examining at least one locus for each chromosome. The frequency of allele loss was highest among the glioblastomas, suggesting that genetic alterations accumulate during glial tumor development. The most common genetic alteration detected involved allele losses of chromosome 10 loci; these losses were observed in all glioblastomas and in three of the anaplastic gliomas. In order to delineate which chromosome 10 region or regions were deleted in association with glial tumor development, a deletion mapping analysis was performed, and this revealed the partial loss of chromosome 10 in eight glioblastomas and two of the anaplastic gliomas. Among these cases, three distinct regions of chromosome 10 were indicated as being targeted for deletion: one telomeric region on 10p and both telomeric and centromeric locations on 10q. These data suggest the existence of multiple chromosome 10 tumor suppressor gene loci whose inactivation is involved in the malignant progression of glioma.

Fine mapping of a region of common deletion on chromosome arm 10p in human glioma.

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Allelic loss on chromosome 10 is a frequent event in high grade gliomas. Earlier studies have shown that in most cases a complete copy of chromosome 10 is lost in the tumor. To define more accurately and specifically the region of common deletion on chromosome arm 10p, we have screened a large series of gliomas for allelic losses that exclusively affect this part of the chromosome. Allelic loss profiles were determined for 127 gliomas, including 118 astrocytomas of various malignancy grades. Seventeen tumors displayed loss of part of chromosome 10. In three of these, only chromosome arm 10p sequences were lost. The interval between loci D10S559 and D10S435 in 10p15, with a length of approximately 800 kilobase pairs, was commonly deleted in the latter tumors, suggesting that this region may harbor a tumor suppressor gene important in glioma tumorigenesis. Comparison of the allelic loss profiles in the low and high grade astrocytomas revealed that astrocytoma progression is associated with increased loss of chromosome 10 sequences.

Acquisition of the glioblastoma phenotype during astrocytoma progression is associated with loss of heterozygosity on 10q25-qter.

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Loss of heterozygosity on chromosome 10 (LOH#10) is the most frequent genetic alteration in glioblastomas and occurs in more than 80% of cases. We recently reported that PTEN (MMAC1) on 10q23.3 is mutated in approximately 30% of primary (de novo) glioblastomas but rarely in secondary glioblastomas that progressed from low-grade or anaplastic astrocytomas. Because secondary glioblastomas also show LOH#10, tumor suppressor genes other than PTEN are likely to be involved. We analyzed LOH on chromosomes 10 and 19, using polymorphic microsatellite markers in microdissected foci showing histologically an abrupt transition from low-grade or anaplastic astrocytoma to glioblastoma, suggestive of the emergence of a new tumor clone. When compared to the respective low-grade or anaplastic astrocytoma of the same biopsy, deletions were detected in 7 of 8 glioblastoma foci on 10q25-qter distal to D10S597, covering the DMBT1 and FGFR2 loci. Six of 8 foci showed LOH at one or two flanking markers of PTEN but did not contain PTEN mutations. LOH on 10p and 19q was found in only one case each. These data indicate that acquisition of a highly anaplastic glioblastoma phenotype with marked proliferative activity and lack of glial fibrillary acidic protein expression is associated with loss of a putative tumor suppressor gene on 10q25-qter.

PTEN (MMAC1) mutations are frequent in primary glioblastomas (de novo) but not in secondary glioblastomas.

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Loss of heterozygosity (LOH) on chromosome 10 is the most frequent genetic alteration associated with the evolution of malignant astrocytic tumors and it may involve several loci. The tumor suppressor gene PTEN (MMAC1) on chromosome 10q23 is mutated in approximately 30% of glioblastomas (WHO Grade IV). In this study, we assessed the frequency of PTEN mutations in primary glioblastomas, which developed clinically de novo, and in secondary glioblastomas, which evolved from low-grade (WHO Grade II) or anaplastic astrocytomas (WHO Grade III). Nine of 28 (32%) primary glioblastomas contained a PTEN mutation and an additional case showed a homozygous PTEN deletion. This indicates that after overexpression/amplification of the EGF receptor, loss of PTEN function is the most common alteration in primary glioblastomas. In this series, 5 of 28 (18%) primary glioblastomas showed both a PTEN mutation and EGFR amplification. In contrast, only 1 of 25 (4%) secondary glioblastomas contained a PTEN mutation, and none of them showed a homozygous PTEN deletion. The secondary glioblastoma with a PTEN mutation developed from an anaplastic astrocytoma that already carried the mutation. The observation that secondary glioblastomas have a p53 mutation as a genetic hallmark but rarely contain a PTEN mutation supports the concept that primary and secondary glioblastomas develop differently on a genetic level.

Role of microRNAs Located on Chromosome Arm 10q in Malignant Gliomas.

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Deletions of chromosome arm 10q are found in most glioblastomas and subsets of lower grade gliomas. Mutations in the PTEN gene at 10q23.3 are restricted to less than half of the 10q-deleted gliomas, suggesting additional glioma-associated tumor suppressors on 10q. We investigated 64 astrocytic gliomas of different malignancy grades for aberrant expression of 16 microRNAs (miRNAs) on 10q. Thereby, we identified four miRNAs (miR-107, miR-146b-5p, miR-346, miR-1287-5p) whose expression was frequently down-regulated in anaplastic astrocytomas and/or glioblastomas. DNA methylation analyses revealed 5'-CpG site hypermethylation of miR-346 in more than two-thirds of primary glioblastomas, while aberrant 5'-CpG site methylation of miR-146b-5p was frequent in IDH1-mutant astrocytomas and secondary glioblastomas. Overexpression of either of the four miRNAs in glioma cell lines reduced cell proliferation and/or increased caspase-3/7 activity. Expression analyses of miRNA overexpressing glioma cells and 3'-untranslated region luciferase reporter gene assays revealed evidence that these miRNAs post-transcriptionally regulate expression of glioma-relevant genes, including CDK6 (miR-107), EGFR (miR-146b-5p, miR-1287-5p), TERT and SEMA6A (miR-346), all of which are overexpressed in malignant gliomas in situ. In summary, we show that the 10q-located miRNAs miR-107, miR-146b-5p, miR-346 and miR-1287-5p are frequently down-regulated in malignant gliomas and thereby may support overexpression of important glioma growth-promoting genes.

The molecular genetics of central nervous system tumors.

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Over the past few years, although much has been learned about the molecular genetics of central nervous system (CNS) tumors, researchers and pathologists are only beginning to understand the scientific basis of the development of these tumors. Data accumulated so far support the division of glioblastoma into two clinical and molecular subsets. Primary or de novo glioblastomas occur in older patients, are clinically aggressive and exhibit epidermal growth factor receptor amplification or overexpression. Secondary glioblastomas develop from pre-existing low-grade astrocytomas, have a more protracted clinical course, and frequently contain p53 mutations. Both types of tumors show deletions of chromosome 10 and possibly mutations of the PTEN/MMAC1 gene as an endstage event. Oligodendrogliomas have been shown to have genetic abnormalities distinct from those of the astrocytic tumors, commonly involving chromosomes 1p and 19q. As regards meningiomas, loss of chromosome 22q and mutations of the neurofibromatosis type 2 gene are frequent events and loss of chromosome 14q and 10q may be seen in atypical or malignant transformation. Such genetic findings, apart from providing a better understanding of neoplastic transformation in brain tumors, are beginning to form the basis of a new approach to neuro-oncology.

Fluorescence in situ hybridization study of aneuploidy of chromosomes 7, 10, X, and Y in primary and secondary glioblastomas.

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The aneuploidy of autosomes 7, 10, and sex chromosomes (X and Y) was analyzed in a series of 44 primary (de novo) and 20 secondary glioblastomas using fluorescence in situ hybridization (FISH) on smear preparations of glioma tissue. The tumors were screened for trisomy 7, monosomy 10, as well as loss of the Y chromosome and disomy of the X chromosome in male subjects, and monosomy of the X chromosome in female subjects. We found that taken alone or in combination, these chromosomal abnormalities do not appear to be characteristic of a glioblastoma subtype; therefore, they do not allow the differentiation between primary and secondary glioblastomas. Also, the loss of a chromosome 10 appears to be an earlier event than a gain of a chromosome 7 for the genesis of a secondary glioblastoma.

Chromosomal abnormalities in human glioblastomas: gain in chromosome 7p correlating with loss in chromosome 10q.

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Various genomic alterations have been detected in glioblastoma. Chromosome 7p, with the epidermal growth factor receptor locus, together with chromosome 10q, with the phosphatase and tensin homologue deleted in chromosome 10 and deleted in malignant brain tumors-1 loci, and chromosome 9p, with the cyclin-dependent kinase inhibitor 2A locus, are among the most frequently damaged chromosomal regions in glioblastoma. In this study, we evaluated the genetic status of 32 glioblastomas by comparative genomic hybridization; the sensitivity of comparative genomic hybridization versus differential polymerase chain reaction to detect deletions at the phosphatase and tensin homologue deleted in chromosome 10, deleted in malignant brain tumors-1, and cyclin-dependent kinase inhibitor 2A loci and amplifications at the cyclin-dependent kinase 4 locus; the frequency of genetic lesions (gain or loss) at 16 different selected loci (including oncogenes, tumor-suppressor genes, and proliferation markers) mapping on 13 different chromosomes; and the possible existence of a statistical association between any pair of molecular markers studied, to subdivide the glioblastoma entity molecularly. Comparative genomic hybridization showed that the most frequent region of gain was chromosome 7p, whereas the most frequent losses occurred on chromosomes 10q and 13q. The only statistically significant association was found for 7p gain and 10q loss.

Loss of heterozygosity for alleles on chromosome 10 in human brain tumours.

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We analysed for loss of alleles on chromosome 10, 25 astrocytomas, 3 ependymomas, 2 medulloblastomas, 2 juvenile pilocytic astrocytomas, 2 gangliogliomas, 1 subependymal giant cell astrocytoma and 1 anaplastic oligoastrocytoma. A battery of 12 DNA markers spanning chromosome 10 was employed. Loss of heterozygosity on chromosome 10 was seen in 16 tumours (13 glioblastoma multiforme, 2 anaplastic astrocytomas, and 1 anaplastic oligoastrocytoma), but not in any of the low-grade astrocytomas examined. High-resolution restriction fragment length polymorphism (RFLP) analysis showed that the loss of alleles in a number of tumours involved two separate large regions of chromosome 10 (10p-proximal 10q and distal 10q). However, a small common region of deletion overlap could not be identified. Our data indicate that the loss of alleles on chromosome 10 is a common finding, seen in over two-thirds of malignant astrocytomas, and may be suggestive of the presence of two or more chromosome 10 tumour suppressor genes involved in astrocytoma formation. Nevertheless, the possibility of these genetic changes being secondary and not causative of the deregulated cell growth cannot be excluded. Regardless of the mechanisms involved, however, chromosome 10 deletions may be a genetic marker for malignant astrocytomas.

A novel gene, LGI1, from 10q24 is rearranged and downregulated in malignant brain tumors.

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Loss of heterozygosity for 10q23-26 is seen in over 80% of glioblastoma multiforme tumors. We have used a positional cloning strategy to isolate a novel gene, LGI1 (Leucine-rich gene-Glioma Inactivated), which is rearranged as a result of the t(10;19)(q24;q13) balanced translocation in the T98G glioblastoma cell line lacking any normal chromosome 10. Rearrangement of the LGI1 gene was also detected in the

A172 glioblastoma cell line and several glioblastoma tumors. These rearrangements lead to a complete absence of LGI1 expression in glioblastoma cells. The LGI1 gene encodes a protein with a calculated molecular mass of 60 kD and contains 3.5 leucine-rich repeats (LRR) with conserved flanking sequences. In the LRR domain, LGI1 has the highest homology with a number of transmembrane and extracellular proteins which function as receptors and adhesion proteins. LGI1 is predominantly expressed in neural tissues, especially in brain; its expression is reduced in low grade brain tumors and it is significantly reduced or absent in malignant gliomas. Its localization to the 10q24 region, and rearrangements or inactivation in malignant brain tumors, suggest that LGI1 is a candidate tumor suppressor gene involved in progression of glial tumors.